

Fixed Partial Denture

Chapter One: Introduction to FPD

- FPD is a partial denture that is cemented to natural teeth or roots which furnish the primary support to the prosthesis".
- Its focus is to restore function, esthetics, and comfort.

Common Terminologies Used in FPD

Crown

- It is a cemented extracoronal restoration that covers or veneers the outer surface of the clinical crown.
- Primary function of a crown is
 - To protect the underlying tooth structure and
 - Restore the function, form and aesthetics.
- **Clinical crown:** is intraorally visible tooth structure.
- **Anatomical crown:** the area of the tooth covered by enamel.
- **Full veneer crown (FVC):** If the prosthetic crown covers all five surfaces of the clinical crown.
- **Partial Veneer Crown (PVC):** If the prosthetic crown does not cover the entire clinical crown.
- **Retainer:** A crown that is used as a part of the FPD for retention and support from the abutment tooth.

Facial Veneers

- Are ceramic prosthesis that are used as a thin layer over the facial surface of the tooth, **primarily for aesthetic** reasons.

Inlay

- It is an intracoronal restoration, which is used for medium sized single tooth **proximo-occlusal** and **gingival** lesions.
- Usually made of gold alloy or ceramic material.

Onlay

- It is an intracoronal restoration, which is used to restore a more extensively damaged posterior tooth with a wide mesio-occluso-distal lesion.
- It is as an inlay supplemented (fused) with an occlusal veneer in order to restore large lesions.

Abutment

- It is any tooth, root or implant which, gives attachment and support to the FPD.

Pontic

- The artificial tooth that replaces a missing tooth in a FPD is called a pontic.
- Pontics are attached to the **retainers**. All forces acting on the pontic are transferred to the abutment through the retainers.

Retainer:

- A crown that is used as a part of the FPD for retention and support from the abutment tooth.

Connectors

- Connection that exists between the pontic and retainer.
- They may be rigid or non-rigid.
 - Rigid connectors are immovable attachments. E.g. Solder joints.
 - Non-rigid connectors are movable attachments with a key-keyway mechanism. E.g. Precision attachments (stress breakers)

Indication for FPD

- Short span edentulous arches
- Presence of sound teeth adjacent to the edentulous space.
- Ridge resorption where a RPD cannot be stable or retentive.
- Patient's preference
- Mentally compromised and physically handicapped patients

Contraindication for FPD

- Large amount of bone loss as in trauma.
- Very young patients where teeth have large pulp chambers.
- Periodontally compromised abutments.
- Distal extension denture bases as in class I and II cases.
- Long span edentulous spaces.
- Bilateral edentulous spaces, which require cross arch stabilization.
- Congenitally malformed teeth, with inadequate tooth structure to offer support.

- Medically compromised patients (e.g. leukemia, hypertension).
- Very old patients.

Diagnosis & Treatment Planning

Diagnosis

- History, clinical examination, radiological examination and diagnostic casts are vital.

History (Psychological and medical conditions)

- **Diabetes:** very high risk of periodontal lesion resulting abutment failure. So Patients should be made aware to control the condition.
- **Xerostomia:** predisposes to caries due to accumulation of debris and decreased buffering capacity.
 - In cases with Xerostomia (Vitamin A deficiency, salivary gland disease) use of anticholinergics, anorectics and antihypertensives should be controlled or avoided and alternate medication should be chosen.
- **Cardiovascular diseases:** in patients using pace makers all electrosurgical procedures are contraindicated for these patients.
 - Adrenaline is avoided in the local anaesthetic and the gingival retraction cord should be free of adrenaline.
- **Miscellaneous conditions:** Drug allergy, nickel sensitivity and presence of any infectious diseases to prevent cross infections.

Clinical Examination

- Extra-oral Examination
 - Any pain, tenderness, clicking sound in the **TMJ**
 - Anatomy, physiology and pathology **muscles of mastication**
- Intra-oral Examination
 - Oral hygiene
 - Presence of attached gingiva
 - Occlusal disharmony and type of occlusion
 - Caries and periodontitis
 - Amount of residual ridge and presence of wear facets.
 - Contour of the ridge and texture of the soft tissues

Radiographic Examination (To detect)

- The number, size and location of caries.
- Caries beneath existing restorations.
- The level of alveolar bone.
- Crown-root ratio of the abutment teeth.
- Morphology of the roots of the abutment teeth.
- Quality of endodontic restorations.
- Width of the PDL space. It is increased in patients with trauma from occlusion.
- Presence of any root stumps in the edentulous area.
- Thickness of the soft tissues in the edentulous area.

Diagnostic Casts of Both Arches

- To assess the dimensions of the edentulous space.
- To observe the height, rotations, inclination of the abutment teeth
- To see the number, size and position of wear facets

- To determine the occlusion and the morphology and also the amount of occlusal load of the opposing teeth

Treatment Plan

- Selection of appropriate prosthesis
 - Depending on factors such as aesthetics, periodontal and biomechanical factors.
- Type of material/technique to be used
- Evaluation of abutment and residual ridge
- Biomechanical considerations
- The patient's needs
- Occlusion with the opposing teeth.

Classification of FPD

- 3 Class, 3 Division, 5 Sub-division and 2 Groups

Class

- Identifies the location of the edentulous space.
- **Class I:** Posterior edentulous spaces. One or more of the posterior teeth are missing.
- **Class II:** Anterior edentulous spaces. One or more of the anterior teeth are missing
- **Class III:** Antero-posterior edentulous spaces. Some anterior and posterior teeth are missing

Division

- Gives info about the teeth adjacent to the edentulous space.

- **Division I: Cantilever FPDs:** Abutments present only on one side of the edentulous space are capable of taking support.
- **Division II: Conventional FPDs:** Abutments that are capable of taking up occlusal load are present on both sides of the edentulous space.
- **Division III: Pier Abutments.** A single tooth is surrounded by an edentulous space on either side. Two division II FPDs separated by a single tooth, which is capable of providing support is a Division III FPD. Such cases are treated with a single prosthesis.

Sub-Divisions

- Denotes the status of the tooth that is to be used as an abutment.
- **Sub-division I: Ideal abutments:** Healthy teeth, which provide good support
- **Sub-division II: Tilted Abutments:** Either the design of the prosthesis should be modified or the tilt of the abutment should be corrected.
- **Sub-division III: Periodontally weak abutment.** This abutment cannot take up occlusal load as effectively as healthy abutment.
- **Sub-division IV: Extensively damaged abutment.** The abutment has good bone support but require extensive restoration e.g. inlay, onlay, dowel core.
 - The status of the abutment crown determines the type of attachment required for the FPD

- **Sub-division V: Implant abutment.**

Each sub-division can be further grouped into A and B.

- **A:** denotes that the support for one side of the edentulous space is taken from a single abutment.
- **B:** Denotes that the support for one side of the edentulous space is taken from more than one abutment tooth.

Types of Residual Ridge of the Patient

Seibert's Classification of Residual Ridge

- **Class N:** indicates normal Ridges
- **Class I defects:** faciolingual loss of tissue width with normal ridge height
- **Class II defects:** loss of ridge height with normal ridge width
- **Class III defects:** a combination of loss in both dimensions.

Chapter Two: Parts of FPD

- Include Retainers, Pontics and Connectors

Retainers

- Part of a FPD that unites the abutment(s) to the remainder of the restoration and obtain support from the abutment.
- **Major retainers:** are retainers, which cover the entire occluding surface of the tooth. E.g. FVC, PVC crowns.
- **Minor retainers:** are small metallic extensions that are cemented onto the tooth. Eg: inlay, onlay.

Types of Retainers

- Based on tooth coverage:
 - Full veneer crowns
 - Partial veneer crowns
 - Conservative (minimal preparation) retainers
- Based on the Material Being Used (explained above)
 - All metal retainers, Metal ceramic retainers
 - All ceramic retainers, All acrylic retainers
- All retainers **except conservative retainers** require the abutment tooth to be prepared (reduced) to accept them. The amount of required tooth preparation varies according to the type of retainer.

1. Full Veneer Crowns

- Cover all the five surfaces of the abutment.
- Fabricated like a cap and are usually indicated for extensively damaged teeth.
- Are the most retentive and ideal retainers because their design can resist masticatory forces in all directions

2. Partial Veneer Crowns

- Preferred over FVC because they require less tooth reduction. But, they are less retentive
- Pins can be fabricated to fit pin holes created on the tooth for additional retention.

3. Conservative Retainers

- Require minimal tooth reduction e.g. acid etching.
- Cannot accept heavy occlusal load and are primarily indicated for anterior teeth.

- They have small metallic extensions, which are designed to be luted directly onto the lingual surface of the abutment teeth using a resin cement.

Criteria for Selecting Type of Retainers

- Alignment of abutment teeth and retention:
 - Parallel abutments could receive FVC with rigid connectors because parallel alignment helps to provide a single path of insertion.
 - Non-parallel abutments (since a common path of insertion is not possible) require pin retained retainers that don't need to be placed along the long axis of the tooth can be given
- Appearance
 - It is best to retain the facial /buccal surface of the natural tooth as they provide the best aesthetics.
- The Condition of the Abutment Teeth
 - PVCs are preferred for non-carious (healthy) abutments or abutments with large restoration but intact facial/buccal surface
 - Endodontically treated teeth may have to be restored with core/post before designing the retainer
- Cost
 - Anterior restorations are best restored with all ceramic crowns. But they are the most expensive.
 - Metal ceramic crowns could be economically the best replacement for both anterior and posterior teeth.
- Preservation of Tooth Structure

- Buccal/facial surface should be conserved.
- Partial veneer crowns are more conservative than full crowns.
- All ceramic crowns are the least conservative

Pontics

- Part of a FPD that replaces the lost natural tooth, restores function and occupies the space of the missing tooth.
- Standard sized teeth cannot be used as a pontic in fixed partial dentures.

Ideal Requirements of a Pontic

- It should restore the function
- It should provide good aesthetics.
- It should be comfortable
- It should be biocompatible.
- It should be easy to clean
- It should preserve underlying mucosa and bone.

Pontic Design

- There are three important factors that control the design of the pontic.
- Edentulous Space available
 - Orthodontic movement of adjacent teeth.
 - Placement of modified full coverage retainers
- The contour of the residual alveolar ridge.
 - Contour of the ridge and texture of the soft tissues should be observed during intra-oral examination.
 - A smooth rounded ridge is best

- In cases with overhanging hyperplastic tissues, surgical excision is done.
- In case of severe resorption ridge augmentation, tissue grafts, etc is done
- Amount of occlusal load on the pontic
 - The pontic should be a replica of the tooth it replaces even if it was altered due to abrasion, etc.
 - It is not advisable to reduce the buccolingual width of the pontic even in cases with compromised periodontal support.

General Design Considerations for a Pontic

- Each surface should be designed to fulfil their requirements.

Gingival surface

- Highly glazed porcelain is the material of choice
- Tissue contact should be minimal to prevent ulceration

Occlusal surface

- Reducing the height of the occlusal table does not help to reduce the occlusal load on the tooth since the proprioceptive impulses from the abutment teeth control the occlusal load,
- The functional cusps of the occlusal surface of the pontic should not be reduced, to preserve a stable vertical dimension. The functional cusp is the load-bearing cusp of the tooth.

- In the maxillary teeth the buccal cusps provide aesthetics. In the lower teeth the lingual cusps aid to protect the tongue.

Proximal surface

- The proximal surfaces of the teeth should be contoured according to the following factors:
- Vertical clearance should be sufficient to permit physiologic contour of the pontic and allow space for the interproximal tissues.
- Interproximal embrasures should be left open to permit easy cleaning.
- The sizes of the maxillary interproximal embrasures should be reduced for the sake of aesthetics. But sufficient space should be given to avoid impingement to the interdental papilla.
- Wider embrasures are provided for posterior teeth for better cleansing. The mandibular molars are provided with large embrasures for better cleaning and aesthetics is not critical.

Buccal and lingual surfaces

- Designed based on the aesthetic, functional and hygienic requirements.
- Facial surface: aesthetics as the primary concern
- Lingual surface: Hygiene as the primary concern
- The embrasures are wider lingually than facially.
- Lingual surface should be similar to an adjacent teeth from the cusp tip till the height of contour, then it should

recede sharply and concavely from the height of contour towards the facial surface to form a pinpoint contact on the labial surface of the residual alveolar ridge.

Design Considerations for a Posterior Pontic

- All surfaces should be convex, smooth and properly finished.
- Contact with the buccal slope of the ridge should be minimal (pin point) and pressure free.
- The occlusal table should be in harmony with the occlusion of all the other teeth.
- The buccal and lingual shunting mechanism should correspond to that of the adjacent teeth.
- The overall length of the buccal surface should be equal to that of the adjacent abutments or pontics.

Design Considerations for an Anterior Pontic

- Same as above but
- A larger contact with the ridge is provided for a natural look. If a pinpoint contact is given for a case with ridge resorption, unaesthetic black spaces may become visible.
- Lingual contour should be in harmony with that of the adjacent teeth.

Classification of Pontics

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Connectors

- The portion of a fixed partial denture that unites the retainer(s) and pontic(s)

Classification of Connectors

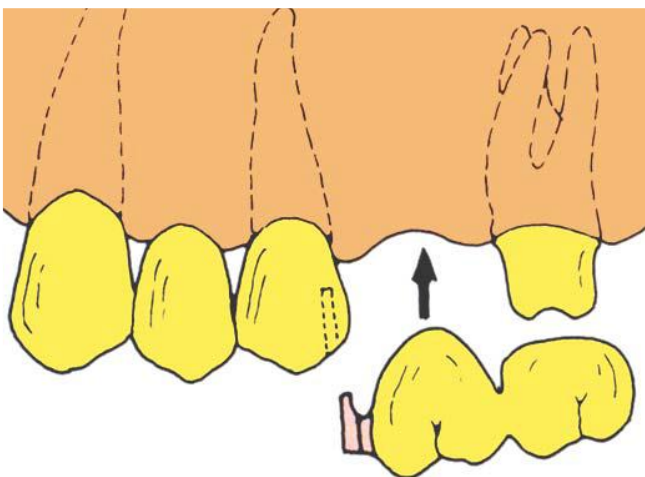
1. Rigid connectors

- Used when the entire load on the pontic is to be transferred directly to the abutments.
- Can be made by casting as part of a multiunit wax pattern or by soldering.

2. Non-rigid connectors

- Indicated in cases where a single path of insertion cannot be achieved due non-parallel abutments.

2.1 Tenon (Male)-Mortise (Female) connectors

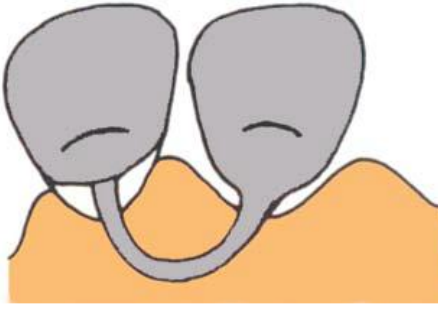


- Is composed of a **key** and **key way**

- The keyway is recessed into the proximal contour of the retainer, and
- The key extends from the pontic

- The distal segment, with the key, is cemented immediately after cementation of the mesial segment

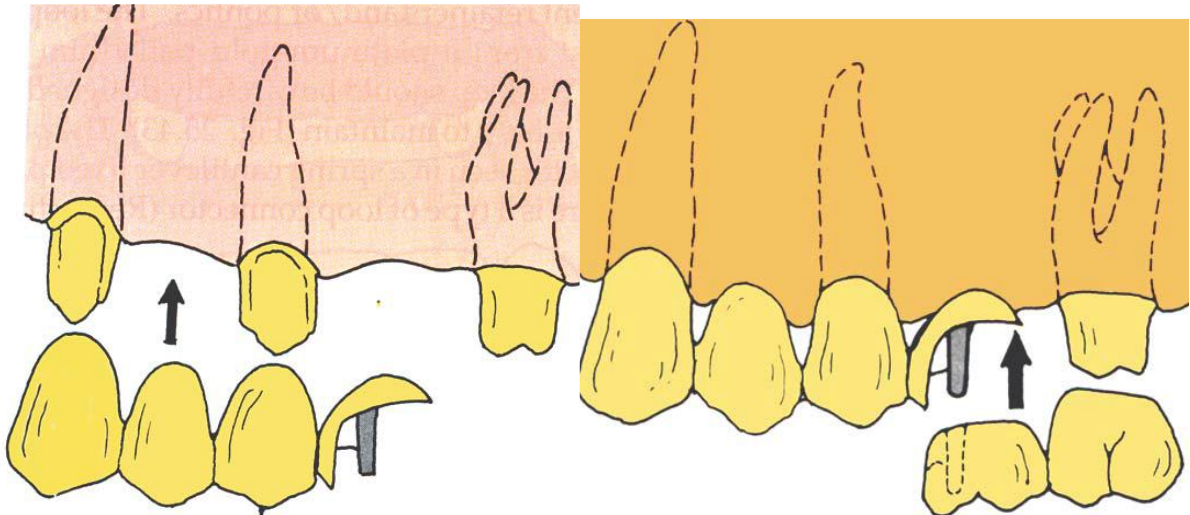
2.2 Loop connectors



- Used when an existing diastema is to be maintained in a planned fixed prosthesis.
- Consists of a loop on the lingual aspect of the prosthesis that connects adjacent retainers and/or pontics.

2.3 Split pontic connectors

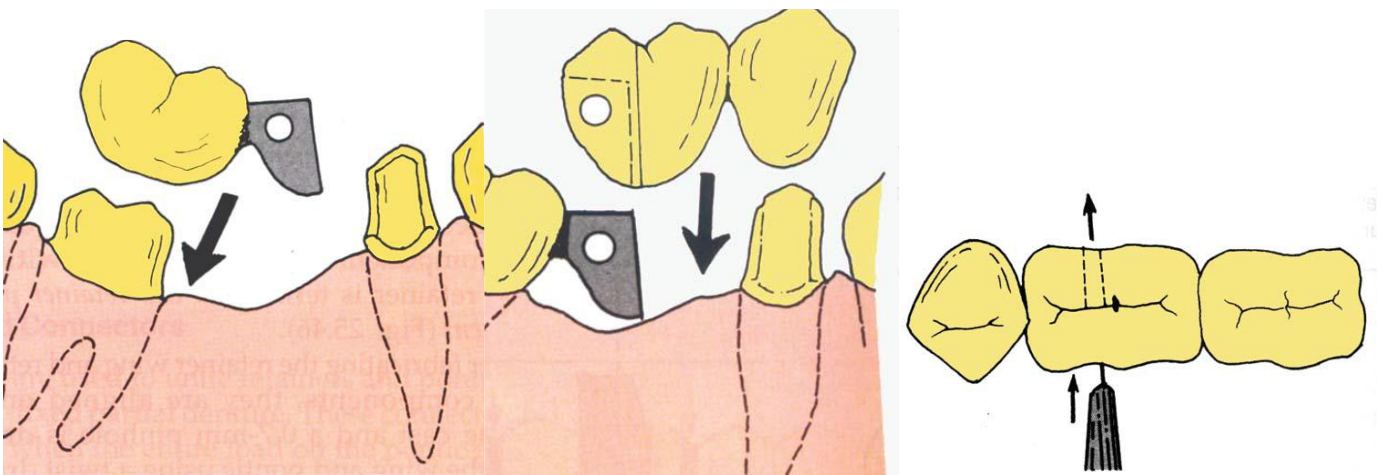
- Used only in cases with a pier abutment
- Here the pontic is split into mesial and distal segments and the connector is incorporated within the pontic.
- Each of these segments are attached to their respective retainers.
- The mesial segment is fabricated with a shoe/key is cemented first. The distal segment is fabricated with a keyway to fit over the shoe.



2.4 Cross pin and wing connectors

- Similar to split pontic connectors and used for tilted abutments.

- A wing is attached to the distal retainer termed **Retainer Wing Component**.
 - The wing should align with the long axis of the mesial abutment.
- The pontic is attached to the mesial retainer and is termed as the **Retainer Pontic Component**.
- The distal retainer and wing should be cemented first and then the mesial retainers with the pontic is cemented. Finally a pinhole is drilled across the wing and pontic then a pin is driven through them.



Chapter 3 Design of FPD

- The major factors to be considered while designing a FPD include:

1. Primary requirement.

- Prosthesis should be designed such that it best fulfills its primary requirement (restore decayed teeth, prevent further decay, to protect weakened, to restore aesthetics)

2. Bio-mechanical considerations.

– I. Length of the edentulous span & Occlusogingival height of the pontic.

- Longer the span, more is flexion of the FPD.

$$\text{Flexion} = \frac{(\text{Length of the Fixed Partial Denture})^3}{(\text{Occlusogingival Height of the Pontic})^3}$$

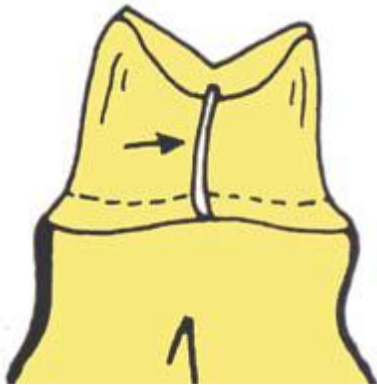
- Increase in length of the span of FPD 2X increases its flexibility by 8X.
- Hence, the flexion of a long span fixed partial denture can be decreased by **increasing the occlusogingival height** of the pontic or by **using high strength alloys like nickel chromium**.
- **Double abutments** (primary and secondary) should be used to overcome long edentulous spans

– II. Arch curvature.

- Curvature of the arch increases the stress developed within a FPD. If pontic is outside the interabutment axis, it will behave like a lever producing torqueing of the abutment.
- Using double abutment prevents this trauma.
- A **canine replacement FPD is subjected to more forces** because the canine forms the curve of the arch and tends to lie outside the interabutment axis.

– III. The direction of dislodging forces acting on the FPD.

- Usually in a single tooth restoration, the dislodging forces act along the BL direction. But in a FPD, they act in a MD direction.



- To resist these dislodging forces, **grooves** may be used on the buccal and lingual surfaces of the prepared abutment in order to enhance resistance and structural durability of the retainer

3. Type of material/Techniques Used

- depends on the following factors
- Age (Young need stronger materials)
- Occlusal load (FVC for high stress)
- Remaining tooth structure (PVC for teeth with small lesions)
- Oral hygiene (Fluoride releasing materials for patients with high caries index)
- Viscosity of saliva
- Type of opposing teeth (Stronger materials to oppose natural teeth).

We will see the materials used later.

4. Abutment selection.

Should withstand both the forces acting on it, and on the pontic. Factors influencing choice are the flg:

- Location, Position and Condition of the Tooth
 - Adjacent to edentulous space,

- Vital but also endo treated can be used
- Not Pulp capped teeth because they are always under the risk of requiring RCT.
- Grossly decayed but that can be restored with FVC and with good PDL support.
- Root Configuration
 - Longer, flat (greater BL width) and also with irregular curvatures are preferred.
- Crown-Root Ratio
 - Should be <1 , 1 is acceptable but >1 is unacceptable
- Healthy alveolar bone and periodontal support
 - **Ante's Law:** The sum of the pericemental areas of abutment teeth should be equal to or surpass that of the teeth being replaced.
 - In RPD sum of pericemental areas of abutment plus mucosal area of denture base should be equal or greater than pericemental areas of the missing teeth.
 - If pericemental area is not sufficient secondary abutment should be used.

5. Residual ridge of the patient

- Type, architecture and amount of destruction affect the design of the **pontic**.

Described earlier

6. Occlusion with the opposing teeth

- High strength alloys should be used for cases with traumatic bite (excessive biting force).

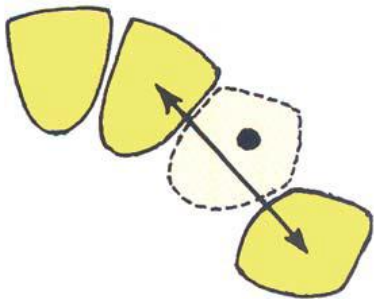
Will be discussed in detail later

Special Cases That Require Unique Design

1. Replacement of a Single Missing Canine

- If cantilever FPD is planned, support should be taken from both CI and LI.

- It is considered as a complex FPD b/c the tooth lies outside the inter-abutment axis and the adjacent teeth like the LI and the 1st PM are very weak.



- This FPD will be subjected to forces acting in an outward direction.

2. Replacement of a Single Missing Molar

- If a cantilever FPD is planned for such cases then support should be taken from both the premolars

3. Tilted Molar Abutments

- Even if it is adjacent to the space we need a single path of insertion.
- This can be accomplished either by up-righting the tooth or by fabricating a partial veneer crown or a telescoping crown.

4. Pier (central) Abutments

- Is central abutment with edentulous space on both sides. In this case, a single prosthesis is fabricated using all three abutments.
- It is subjected to leverage and torsional forces acting on the prosthesis on either side.
- To protect it specially designed FPD with non-rigid connectors are used.

Chapter 4: Occlusion in FPD

Occlusion

- The static relationship between the incising and masticating surfaces of the maxillary or mandibular teeth or tooth analogues.

Normal Occlusion (Angles class I)

- The mesiobuccal cups of the maxillary first molar is aligned with the mesiobuccal groove of the mandibular first molar.

Ligaments of the TMJ

- **1. Lateral (Temporomandibular) Ligament**
 - Functions to limit the amount of rotation of the mandible and protects the retrodiscal pad against posterior movement of the condyle.
 - Has superficial oblique portion and a deep horizontal portion
- **2. Sphenomandibular Ligament**
 - Limit separation between the condylar process and the disc.

– 3. Stylomandibular Ligament

- It limits protrusive movement of the mandible.
- It is a thickening of the deep cervical fascia.

Ligaments of the TMJ			
	Origin	Insertion	Function
Lateral			
Sup Oblique	Articular eminence	Posterior to condylar neck	Limits mandibular rotation
Deep Horizontal			
Sphenomandibular			
Stylomandibular			

Muscles of Mastication

- The muscles of mastication and supra-hyoid muscles control the movement of the mandible and hence the condyle.
- **Temporalis**: Posterior fibres produce retraction while the anterior and middle fibres produce elevation of the mandible.
- **Masseter**: Elevation and lateral movement of the mandible.
- **Medial pterygoid**: Elevation and lateral movement of the mandible.
- **Inferior head of lateral pterygoid**: Depression, Protrusion and lateral movement.
- **Superior head of lateral pterygoid**: It positions the disc during closure.

- **Suprahyoid muscles:** Especially the digastric and mylohyoid. They elevate the hyoid bone or depress the mandible.

Mandibular Movement

- Is a complex 3D movement & described as projected in three perpendicular planes, namely, sagittal, horizontal and frontal.
- This movement comprises of two basic components:
 - 1. **Translation:** When all points within a body have identical motion
 - 2. **Rotation:** When the body is turning about an axis.

1. Sagittal Plane

- Mandible is capable of **pure rotational** and **translational** movement.
- Rotation is limited to about 20-22 mm of incisor separation after which the mandible begins to show translation.
- Rotation occurs around the terminal hinge axis, which is an imaginary horizontal line through the rotational centres of the left and right condyles.

2. Horizontal Plane

- Mandible is capable of rotation around several vertical axes.
- For e.g. during lateral movement, there is rotation around an axis situated in the working condyle along with a slight lateral translation known as "Bennett Movement".

- The non-working condyle travels forward and medially as limited by the glenoid fossa.

3. Frontal Plane

- When a lateral movement is observed in the frontal plane, the working condyle rotates around a sagittal axis perpendicular to this plane. The non-working condyle moves downwards and medially.

Factors Affecting Mandibular Movement

- TMJ
- Teeth influence (occlusion)
- Neuromuscular coordination
- Hinge axis
- Muscles and ligaments

Determinants of Mandibular Movement

- Anatomic structures that dictate or limit movements of mandible.
- **Anterior determinants**
 - Overbite and overjet and palatal surfaces of maxillary anteriors (Incisal guidance).
- **Posterior determinant**
 - Condylar guidance. (Slope of the articular eminence).
- **Proximity of the cusp** to the controlling factors.
 - E.g. posterior occlusal relationships are determined more by condylar guidance than the incisal guidance.
- **Vertical Determinants**
 - Heights of the cusp and depths of the fossa

– Horizontal Determinants

- Relationships that influence the direction of ridges and grooves on the occlusal surfaces

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Chapter 5: Type of Abutments

Healthy/Ideal Abutments

- An **un-restored vital tooth** in its **normal anatomic position** is considered as an ideal abutment with
 - Ideal crown root ratio.
 - Adequate thickness of enamel and dentin.
 - Adequate bone support
 - Absence of periodontal disease
 - Proper gingival contour

Cantilever Abutments

The requisites for a cantilever abutment are:

- More than average bone support
- Sufficient amount of tooth structure
- Non-Endodontically treated
- Teeth with short roots are contraindicated.
- Certain ideal cantilever situations include:

- Replacement of lateral incisor with canine support.
- Replacement of first premolar with second premolar and first molar support.
- If strong abutments are not available, spring cantilever designs can be incorporated.

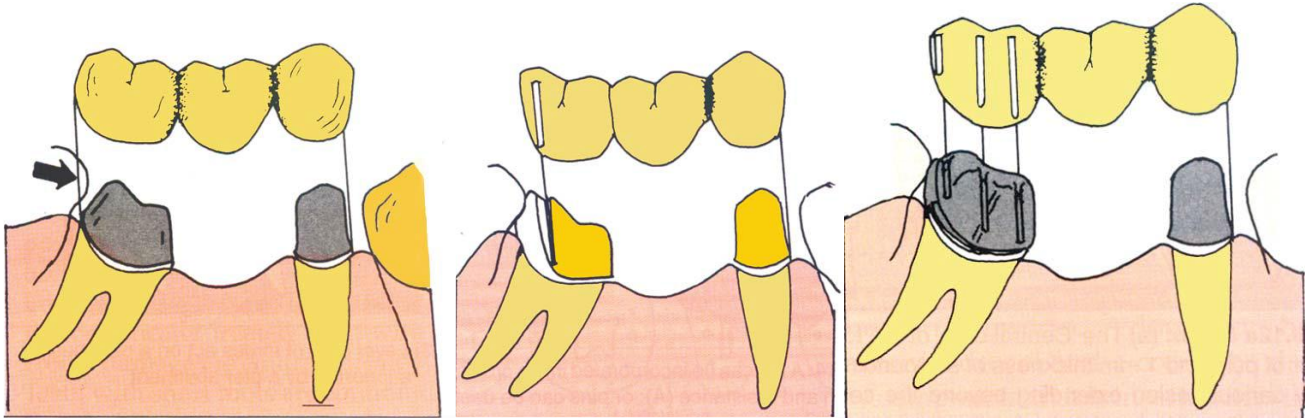
Pier Abutments

- Since a single abutment supports two edentulous spaces, it will be subjected to unbalanced forces, which can lead to trauma of the periodontium.
- The forces acting on one end tend to lift the other end like a lever using the abutment as a fulcrum.
 - To prevent this a stress breaker (non-rigid connector with a key in a keyway) is used on the pier abutment.
- A non-rigid connector is avoided when the posterior abutment has no antagonist. This is b/c supra-eruption of the posterior abutment will occur and unseat the key.

Tilted Abutments

- Single path of insertion is difficult in this case.
- To overcome the disadvantages of a tilted abutment:
 - Mesial Reduction of the abutment
 - Modified Partial Veneer Crown prepared such that the mesial undercut is removed.
 - Orthodontic treatment to upright the abutment
 - Using non-rigid connector
 - Telescopic Crown
 - The tooth is reduced considerably. A coping or retainer is fabricated over the tooth so that it

alters the contour of the crown. This crown should be fabricated with vertical slots so that it can receive a second crown (telescopic crown) in a vertical direction



Extensively Damaged Abutments

- The teeth should be restored such that they gain adequate retention and resistance to be used as an abutment.
- Superficial multiple carious lesions involving all the surfaces require a more extensive restoration than a tooth with a deep carious lesion involving more than one surface.

Precautions during Restorations

- The tooth should not be reduced beyond the central core (the **central core** is a cylinder of pulp with 1mm of surrounding dentin).
- For all restorations except full veneer crowns, all the tooth surfaces should be uniformly reduced.
- If the caries extends to the central core then it should be scooped out and GIC restoration should be done.
- Retentive grooves should not extend more than 1.5 mm below the cervical line and should be at least 1mm wide.

Grooves

- To provide retention
- Should be at least 1mm wide and 1.5mm below the cervical line.

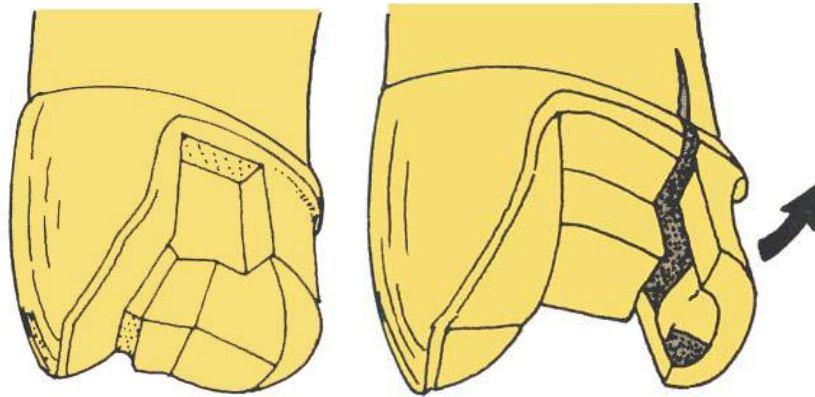
Pins

- It is part of the cast restoration which would seat into these pinholes.
- Used to augment retention and resistance
- Does not require vertical supragingival tooth structure. It can even be used in teeth with inadequate axial wall height.
- Types of Pin Retention
 - Parallel pinholes screwed into the dentin
 - Non-parallel pins screwed into the dentin
- Guidelines to drill a pin hole:
 - It Should be drilled into sound dentin
 - It should not undermine the enamel
 - They should not enter the PDL
 - Pins should not encroach the pulp.
 - The most appropriate points for placement of pins on a tooth are the four corners (MB, DB, DL, ML)

Box Forms

- A retentive box can be used if a proximal lesion on the tooth interferes with the placement of a retentive groove.
- Functions to provide both retention and resistance.
- Proximal boxes should not be placed within 180° the tooth circumference as this would weaken the tooth structure and lead to fracture.

- In such cases a pin retained amalgam core can be prepared and a FVC should be placed over it.



Restoration of Endodontically Treated Teeth

- To use FVC in endo treated tooth a dowel core (Post Core) is used to increase its resistance and support the crown.
- A dowel core has two parts namely a dowel and a core.
 - Dowel is the screw component that is inserted into the root canal and
 - Core is the retentive component, which behaves like a prepared crown for the placement of retainer.
- Dowel Core Preparation?????
- Types of Dowel Cores ?????

Implant Abutments

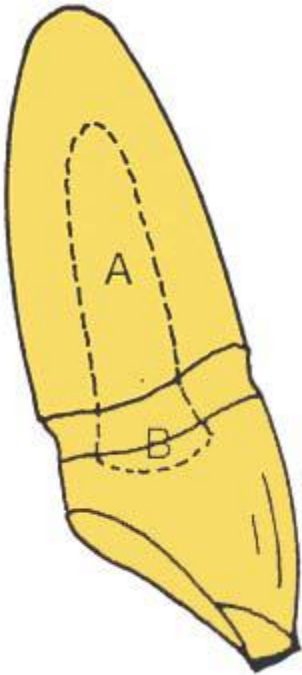
Indication

- Partially edentulous patient who is unable to wear a RPD or CD.
- Long-span fixed partial dentures with poor prognosis.
- Unfavourable number and location of potential tooth abutments.

- Unfavourable number and location of potential tooth abutments.
- Single tooth loss that would necessitate preparation of undamaged teeth for a fixed prosthesis.
 - E.g. missing single lateral incisors.

Contraindication

- Acute and Terminal illness.
- Pregnancy.
- Uncontrolled metabolic disease.
- Tumoricidal radiation given at the implant site.



- More on Implants????????? ???????
 ???

Chapter 6: Tooth Preparation

Principles of Tooth Preparation

1. Preservation of tooth structure

- There should be minimal possible reduction done.

2. Retention and resistance

Retention:

- The ability of the preparation to prevent displacement of the restoration in a direction opposite to the path of insertion.
- One of the major sources of retention in a fixed partial denture is **tensofriction**.

Retention can be of classified as follows:

- **Primary Retention:**
 - **Sleeve retention** provided by the opposing vertical surfaces of the tooth preparation.
 - **Wedge type retention** seen in intracoronal restorations.
- **Secondary retention:**
 - Retention obtained by retentive features like pins, boxes and grooves, etc.

Resistance

- Is the ability of the prosthesis to resist displacement by forces directed in an apical or oblique direction

In order to create balance between retention and resistance forms; flg features should b

–

3. Structural durability

–

4. Marginal integrity

—

5. Preservation of Periodontium

—

Types of Materials Used in FPD

Plastic materials: e.g. Resin

- Require less amount of tooth preparation. But their success is limited due to their poor strength.
 - Used as temporary crowns over the prepared teeth.

Cast metal:

- Strong and provides a very good fit and finish. Excellent success rate and hence, are most commonly used.
- Two Types: intra and extracoronal replacements.
- **Intracoronal replacements:** Gold is most commonly used. But with a major disadvantage requiring extensive tooth preparation even for small lesions.
 - Base metal alloys are widely used.
- **Extracoronal replacements:** used for teeth with extensive carious lesions. It requires extensive tooth preparation.
 - The margins are placed near the gingiva, predisposing it to periodontal problems.

Metal ceramics:

- Require extensive tooth reduction.
- The junction between the metal and the ceramic is very unaesthetic.
- For aesthetic reasons the margins of the restoration is placed subgingivally which frequently leads to periodontal problems.

Resin veneer:

- It is used only for long-term temporization.

Fiber reinforced resin:

- Type of fiber reinforced indirect composite.
- Provides a good fit and finish and is very aesthetic.

All ceramic:

- It has the best aesthetics but worst marginal finish.
- It is a brittle material due to low elastic strain. Now high strength fillers like alumina and zirconia are used to improve the tensile strength.

All Acrylic:

Used for long-term temporary FPD. They are not indicated for permanent restorations

- Conventional tooth supported fixed partial dentures:
- (Fig. 24.8) It is indicated for the following cases.
 - • Patients with periodontally sound teeth.
 - • Short and straight edentulous span.
 - • Absence of any uncorrectable soft tissue defect.
 - • Presence of proper salivation (fluoride application is advised for xerostomic patients).
-
- Resin bonded fixed partial denture: (Fig. 24.9) It is indicated in:
 - • Presence of defect-free abutments.
 - • Presence of single missing anterior tooth or premolar.
 - • Sometimes a single missing molar with minimal opposing occlusal load.
 - • Presence of sound abutments on either side of the edentulous space.
- Young patients with low-risk of caries and large pulp chambers.
- Abutments with less than 15° axial inclination.
- Absence of deep vertical overlap.
-
- **Implant supported fixed partial denture** (Fig. 24.10)
- It is indicated in

- • Absence of sufficient number of abutments
- for a long edentulous span.
- • Absence of distal abutment but presence of
- good density of bone.
- • Broad flat ridge configurations.
- Single tooth replacements.
- • High caries risk patients.
- • Young adults.